

Characteristic-Time Analysis: Retained Mass versus Microtubule Count

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Parameters

Common to every run in both tables:

Advective velocity	$v = 10^4$
Switch rates	$a = b = w = 100$
Dimensionless solution time	$T = 1$
Domain radius	$R = 1$
Diffusion coefficient	$D = 1$
Microtubule extraction width	$d_{\text{tube}} = 0$
Initial condition	centred

N denotes the number of evenly spaced microtubules, t^* the characteristic time, and m^* the retained mass evaluated at t^* .

Domain grid: 96 x 96

N	t^*	m^*	Wall time (hr)
4	0.073186	0.057734	10.61
8	0.067073	0.147390	10.72
16	0.059530	0.377297	10.88
24	0.054723	0.578565	11.09

Domain grid: 48 x 48

N	t^*	m^*	Wall time (min)
4	0.070148	0.096793	10.5
8	0.061825	0.295470	10.7
16	0.052357	0.670035	11.1
24	0.046611	0.858021	11.6